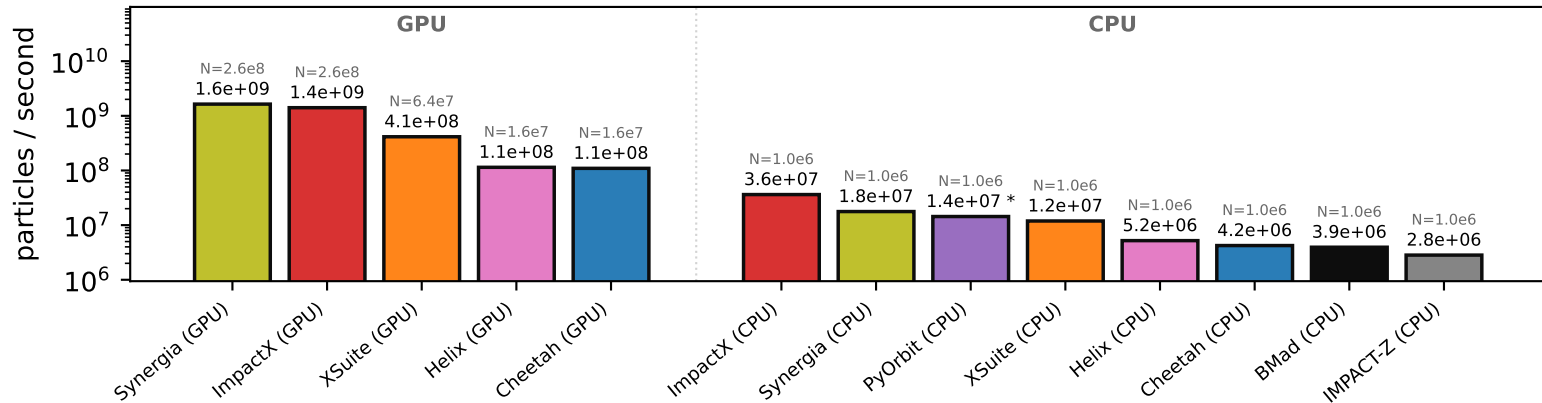


Perlmutter · 3D space charge — best per code, GPU vs. CPU — ref: impactx



* uses a cheaper, less-rigorous Green function (point-1/r vs the integrated GF)

each bar = the fastest config for that code on that device (any precision / fast-math); N = the particle count that bar was measured at (top of each device's sweep)

hardware: CPU: AMD EPYC 7763 (64 cores) GPU: NVIDIA A100-SXM4-40GB

versions: synergia v2024.04-209-g795b99f39 · impactx 26.08 · xsuite 0.53.4 · helix v1.10.1 · cheetah 0.8.2 · pyorbit 96c8def · bmad 20260904-1 · impactz V2.7.7